



School of Risk and Actuarial Studies

ACTL4305/ACTL5305

Actuarial Data Science Applications

Industry Datathon Challenge 2026

This is a real industry project (not just another assignment)!

Insurance Pricing Promises & Transparency

1. Scenario

You are the pricing team at **Saltbush Direct**, a mid-sized Australian general insurer writing comprehensive and third-party motor insurance. The Chief Actuary has asked your team to design the rating structure and pricing approach for the **2027 and 2028 renewal cycles**.

Two things are on your desk before you start:

1. ASIC's [Report 765 When the price is not right: Making good on insurance pricing promises](#) (June 2023), which found that general insurers had overcharged more than 5.6 million consumers by over \$815 million because pricing promises, such as no-claims discounts, loyalty discounts, and multi-policy discounts, were not delivered in full.
2. ASIC's [Report 838 Road testing transparency in car insurance premiums](#) (August 2026), which found that no insurer in its sample explained in quote or renewal documents the key factors driving a customer's premium or why it had changed, and that 72% of consumers who recalled a reason for a price rise were only given a generic explanation.

"I don't want us in the next version of either of these reports. Build me a rating structure that's actuarially sound, commercially workable, and that we can actually explain to a customer without a regulator finding a gap between what we promised and what we delivered."

— The Chief Actuary

2. Learning objectives

By completing this assignment you will:

- Build and validate a claims frequency/severity rating structure from a realistic policy-level dataset, using a modelling method from the course.
- Translate a technical (risk-based) price into a **commercial price** using standard market practices (price floors/ceilings and discount promises) without breaking the promises you make.
- Diagnose the specific pricing-engine design fault (order of operations) that caused nearly half of the \$815 million in ASIC-identified consumer harm, and demonstrate your rating structure avoids it.
- Draft a consumer-facing renewal disclosure that would meet the standard ASIC is now holding insurers to, and defend your design choices against specific findings in REP 838.
- Reflect on the product governance controls a real insurer would need to keep delivering on these promises at scale, year after year.

3. Background you need from the two reports

You do not need to read either report in full, but your report should show you understand the following concepts (all defined and illustrated in REP 765 and REP 838, cited where relevant):

- **Cupping (or collaring) and capping.** Cupping/collaring applies a price *floor* to limit how much a renewing customer's premium can fall year-on-year. Capping applies a price *ceiling* to limit how much it can rise. Insurers use these to manage volatility and retain market share, but they can silently swallow a promised discount if applied in the wrong order (REP 765, pp. 17-18, where the worked example with a \$100 pre-discount premium, a 10% discount, and a \$95 floor is a good example for Task 2).
- **No-claims discount (NCD) / loyalty discount / multi-policy discount.** Each is a form of "pricing promise" under ASIC's definition, meaning a representation that a customer will pay less, or receive a benefit, for meeting some criterion. REP 765 found insurers routinely failed to deliver these in full, and separately raised the risk of a "**loyalty tax**", charging renewing, price-inelastic customers more than an equivalent new customer, which can offset or reverse a promised loyalty benefit (REP 765, pp. 21-22).
- **Dollar disclosure and premium-change explanation.** REP 838 found that even where insurers disclosed premiums prominently, none explained the factors behind the calculation or a specific year-on-year change, and most instalment-vs-annual cost differences were not clearly quantified in dollar terms.

4. The dataset

You have been given a simulated policy-level dataset. The three files are:

- **claims_history_2023_2026.csv.** A policy-year panel (~74,500 rows, ~26,000 policies) with rating factors and claim outcomes, for fitting your model(s).
- **renewal_book_2027.csv.** ~17,200 policies due for renewal on 1 January 2027, with **Premium_2026**, the amount each policyholder actually paid last year. **This is the book you price in Task 2.**
- **renewal_book_2028.csv.** The same policyholders one year later (after attrition), with 2027 claims experience folded into an updated NCD level, plus ~1,520 new-business policies. There is **no**

Premium_2027 column, so you must construct the appropriate prior-year baseline for the 2028 capping/cupping rules from your own Task 2 output. **This is the book you price in Task 3.**

Potential rating factors are chosen to reflect factors commonly used or disclosed by Australian motor insurers: driver age and licence history, vehicle age/value/type, a postcode-level risk zone and its population density, annual kilometres, excess, where the vehicle is garaged overnight, whether it's used for business, payment frequency, no-claim discount level, tenure, and multi-policy status. Deciding what to keep is part of Task 1.

4.1 Data variables

Variables in all three files

Variable	Type	Description
PolicyID	string	Unique policy identifier. Join key across files. New business added for 2028 uses a NEW##### prefix
DriverAge	int	Age of the rated driver in that year
YearsLicensed	int	Years the driver has held a licence
VehAge	int	Vehicle age in years
VehValue	numeric (\$)	Insured (agreed/market) value
VehicleType	factor	Sedan / Hatch / SUV / Ute / Sports
RiskZone	factor	Postcode-level risk zone, A_Metro (highest) to E_RemoteRural (lowest), modelled on the ABS remoteness structure
Density	numeric	Approximate population density of the postcode area (people/km ²). Correlated with, but not identical to, RiskZone
AnnualKms	factor	Low / Medium / High annual driving distance band
ExcessChosen	int (\$)	Voluntary excess selected: 500 / 750 / 1000
GaragingLocation	factor	Garage / Carport / Street (where the vehicle is kept overnight)
VehicleUse	factor	Private / Private_Business (whether the vehicle is used for business as well as private purposes)
NCDLevel	int (0-6)	No-claim discount level, 6 = maximum
TenureYears	int	Years continuously insured with this insurer
MultiPolicyFlag	0/1	Holds a home/contents policy with the same insurer
PaymentFrequency	factor	Annual / Monthly. Affects claim frequency in this dataset and should be included in your Task 1 model, as well as considered in your Task 2 commercial pricing

History-only variables (claims_history_2023_2026.csv)

Variable	Type	Description
Year	int	Policy year, 2023-2026
Exposure	numeric	Fraction of the year on risk (1.0 for every row in this file)
ClaimNb	int	Number of claims in that policy-year
ClaimAmount	numeric (\$)	Total incurred claim cost for that policy-year (0 if ClaimNb = 0)

Renewal-book variables (*renewal_book_2027.csv* / *renewal_book_2028.csv*)

Variable	Appears in	Type	Description
Premium_2026	2027 book only	numeric (\$)	The premium this policyholder actually paid for their 2026 policy year, produced by the legacy pricing model in §4.2 (your prior-year baseline for capping/cupping)
Claims2027Count	2028 book only	int (or NA)	Claims incurred during the 2027 policy year. NA for new business
NewBusinessFlag	2028 book only	0/1	1 if this policyholder is new business joining for 2028, with no prior motor premium to compare against.

4.2 The 2026 legacy pricing model

Premium_2026 was not produced by the risk model you're asked to estimate in Task 1. It was produced by Saltbush Direct's outgoing legacy **rating engine** (the system that turns a customer's rating factors into a price), a simpler, coarser model that the business is retiring. That's why you've been asked to build a new one. The legacy model is:

```
LegacyTechnicalPrice = 1000
  x ZoneFactor x VehicleTypeFactor x AgeBandFactor x NCDFactor
  x (VehValue / 24000) ^ 0.30
Premium_2026 = LegacyTechnicalPrice x 1.35    (rounded, floored at $350)
```

Factor	Legacy grouping and multipliers
Zone	A_Metro -> Metro 1.25; B_InnerRegional/C_OuterRegional -> Regional 1.00; D_Rural/E_RemoteRural -> Rural 0.80
Vehicle type	Sedan/Hatch/SUV/Ute -> Standard 1.00; Sports -> Performance 1.45
Age band	Under 25: 1.35; 25-64: 1.00; 65+: 1.15
NCD	1 - 0.04 x NCDLevel (capped at a maximum discount of 24%, i.e. NCDLevel 6)

Note that **DriverAge** and **NCDLevel** in *renewal_book_2027.csv* are the values entering 2027 and may therefore differ from the corresponding values that applied when Premium_2026 was calculated. For the 2027 cap/collar comparison, use the **Premium_2026** value provided in the renewal book.

Notice what the legacy model does *not* use: **AnnualKms**, **ExcessChosen**, **GaragingLocation**, **VehicleUse**, **TenureYears**, **MultiPolicyFlag**, **Density**, or **PaymentFrequency**, and it groups **RiskZone**, **VehicleType**, and age far more coarsely than your Task 1 model will.

5. Tasks

Task 1: Build and validate the rating structure (technical price)

Task 1a: Exploratory data analysis and cleaning

Before modelling, explore and clean *claims_history_2023_2026.csv*. The data, as provided, is not analysis-ready. You must:

- Explore the dataset and profile its variables.
- Identify the data quality issues you encounter.
- State what you found, what you did to resolve each issue, and why.

This step should be documented briefly in your report (it does not need to be exhaustive) and must be reproducible in your submitted code.

Task 1b: Model building and validation

Using your cleaned data, build a frequency-severity (or pure premium) model to produce a **technical price** for any policy in the 2027/2028 books. You must:

- Choose any method covered in the course, and justify your choice against at least one alternative.
- Validate your model appropriately.
- Investigate whether frequency and/or severity trend over **Year** in `claims_history_2023_2026.csv` (which spans four policy years), and state what you find.
- State and justify an explicit assumption (e.g. an extrapolated claims-inflation rate, or a stated decision not to trend) for how you bring your fitted model's output up to the price level of the year you are actually pricing, and apply it consistently in Tasks 2 and 3.

Task 2: Price the 2027 renewal book

Convert your Task 1 technical price into a **commercial price** for every policy in `renewal_book_2027.csv`, applying the pricing steps below in an order you decide and justify (this ordered sequence of steps, from technical price to final price, is called a **pricing waterfall**). Unlike Task 1, several elements of this waterfall are fixed by "the business." Your job is reconciling those *given* commercial promises with *your own* technical relativities.

A reconciliation risk worth watching for. `NCDLevel`, `TenureYears`, and `MultiPolicyFlag` may be retained in your Task 1 technical model. Each of them is also governed by one of the fixed pricing promises below. If your technical price already reflects a customer's NCD, tenure, or multi-policy status, and you then apply the matching promised discount on top of it, that risk factor is being counted twice rather than reconciled once. Working out how to convert a technical price that reflects all your rating factors into a commercial base that leaves room for the business's promised schedule, without double-counting the factors that schedule already covers, is part of this task.

Expense and profit loading

Build your loading from three components:

- A **fixed per-policy cost** of \$85, representing IT/administration costs that do not scale with risk.
- A **variable expense rate** of 10% of the loaded premium, representing claims handling and other costs that scale with premium.
- A **target profit margin** of 8% of the loaded premium.

Combine them using the standard loaded-premium relationship:

$$\text{LoadedPremium} = (\text{TechnicalPrice} + \text{FixedCost}) / (1 - \text{VariableExpenseRate} - \text{ProfitMargin})$$

These are commercial assumptions supplied by Saltbush Direct rather than quantities you are expected to estimate from the claims data.

Fixed pricing promises

The following are commercial decisions already made by Saltbush Direct's product team. They are not something you estimate from the data, and they will not, in general, match the relativities your Task 1 model finds:

Promise	Schedule
No-claims discount, by NCDLevel	0: 0% · 1: 10% · 2: 20% · 3: 30% · 4: 40% · 5: 50% · 6: 60%
Loyalty discount, by TenureYears	0-2 yrs: 0% · 3-5 yrs: 3% · 6-9 yrs: 6% · 10+ yrs: 10%
Multi-policy discount	Flat 10% if MultiPolicyFlag == 1

Apply each as a percentage reduction from the pre-discount base. Part of your task is deciding, and justifying, whether these apply multiplicatively or additively when more than one applies to the same customer. Either is defensible if stated and applied consistently.

Cupping/collaring and capping

You must design both, each relative to **Premium_2026**, and justify each threshold rather than choosing it arbitrarily:

- A **cap** (ceiling) limiting how much a renewing premium can rise year-on-year.
- A **collar** (floor) limiting how much it can fall.

For the cap, Saltbush Direct's management have supplied the following (hypothetical, market-research-based) estimate of *additional*, price-driven lapse risk from a renewal premium increase, on top of the baseline lapse behaviour already reflected in the data. This relationship is *not* something you can validate from **claims_history_2023_2026.csv** or the renewal books. It is a business assumption given to you for this decision.

Renewal premium increase	Additional lapse risk
0-5%	+0%
5-15%	+5%
15-30%	+15%
More than 30%	+35%

Use this table to justify where you set your cap. A full demand/elasticity model is out of scope. Referencing the table to justify your chosen threshold is sufficient.

For the collar, there is no equivalent table. It's your own design choice. Justify it with reference to the loyalty-tax risk discussed in §3.

Instalment loading

For policies with `PaymentFrequency == "Monthly"`, use an instalment loading of 3%, representing genuine transaction/admin costs of collecting premium in instalments (e.g. payment-processing fees).

You must decide, and justify, **the exact order in which all of the above steps are applied**, and demonstrate that your order does not silently reduce or cancel any promised discount, using the worked-example method from REP 765 p.18 (show the pre-discount premium, whether the floor/cap is triggered, and the discount delivered, at every level of the systematic test in Task 2a below).

Task 2a: Systematic promise testing

A pricing structure can look correct for a few examples without actually being correct in general. Design and carry out your own systematic test showing that every promised discount (NCD, loyalty, multi-policy) is delivered in full across its entire eligibility range, through your complete pricing waterfall. You choose how to structure the test, but describe your method clearly enough for someone else to rerun it and get the same result.

Task 2b: Loyalty tax check

Using your full priced 2027 book, test whether, after controlling for risk, longer-tenure customers are paying *more* relative to their technical price than shorter-tenure customers of similar risk. Because the loyalty discount schedule in Task 2 is fixed by the business and independent of whatever tenure effect your Task 1 model actually estimates, your own pricing structure may or may not turn out to create a loyalty tax, depending on how the fixed schedule compares to your model's estimated tenure relativity.

Task 3: Price the 2028 renewal book

Repeat the exercise for `renewal_book_2028.csv`. The prior-year baseline for your capping/cupping rules should be constructed from an appropriate stage of **your own Task 2 pricing output**, joined by `PolicyID`. Consider which stage of the pricing waterfall should be carried forward to avoid double counting when the 2028 pricing steps are applied, and briefly justify your choice.

Handle the following correctly:

- Policies with `NewBusinessFlag == 1` have no prior-year premium and no tenure/loyalty history.
- `NCDLevel` and `TenureYears` have already been updated for 2028. `NCDLevel` reflects 2027 claims experience. Use them as given.
- Refit your Task 1 models on the same 2023-2026 claims history you used for Task 1. `renewal_book_2028.csv` includes `Claims2027Count`, but no equivalent 2027 claim-cost data is provided, so a full model refresh incorporating 2027 experience isn't possible from the data given and isn't expected here. This keeps the exercise focused on the rating-engine change below.

A rating-engine change. From 2028, Saltbush Direct's compliance team has ruled that `PaymentFrequency` can no longer be used as a rating factor. You must:

- In 2-3 sentences, explain why pricing based on how a customer pays might raise fairness or conduct concerns.

- Remove **PaymentFrequency** from your technical price for the 2028 book. You may still apply the separate 3% instalment loading for policies paid monthly, since this reflects transaction/admin costs associated with collecting premium in instalments rather than a risk-based rating factor.
- Comment on the impact. How much did **PaymentFrequency** matter to your Task 1 model, and does removing it force you to revisit your order of operations or any promised discount?

This is similar to how the order-of-operations bug in REP 765 arose in practice, not because the original implementation was wrong, but because a later, unrelated system or product change was layered on top of it without re-checking that promised discounts still survived the full waterfall.

Separately, comment briefly (2-3 sentences) on whether applying the same year-on-year capping/collaring rules for a second consecutive year to the same customers raises any new risk of a compounding loyalty tax, referencing REP 765's discussion of loyalty representations as "a moving target."

Task 4: Communicate the price transparently

Choose **one** customer from your priced 2027 book whose premium changed materially between 2026 and 2027. Draft the premium-related content of a **renewal notice** for that customer (the part of the letter/email dealing with price. You do not need to draft a full Product Disclosure Statement, the longer legal document that covers what's insured and excluded). Use the disclosed 2026 legacy pricing model (§4.2) as your baseline for this comparison. You know exactly which factors it used and how coarsely, so you can attribute the change precisely, rather than guessing at an unknown prior model. It must:

- Display the current and prior-year premium in direct comparison, in dollars, with clear labelling of what the prior-year figure represents.
- Explain the specific, non-generic factors that drove this customer's price and price change.
- State any discount(s) applied, in dollar terms, with a plain description of the eligibility basis.
- If your case-study customer's **PaymentFrequency** is Monthly (or if you want to show them the alternative), state the dollar cost difference between paying annually and by instalment explicitly.
- Be understandable to a policyholder with no insurance or maths background.

Follow this with a **design commentary** (300-400 words) that maps at least four specific choices in your notice to specific findings from REP 765 and/or REP 838 (e.g., "we state the dollar difference for instalments, because REP 838 Finding 6 found most insurers didn't disclose this clearly").

Task 5: Governance reflection

Explain what internal controls a real insurer would need for your rating structure and discount promises in Tasks 2-4 to keep being delivered correctly at scale, over many years and renewal cycles, without the kind of drift ASIC found. Task 2a asked you to prove your *current* implementation delivers every promise correctly. This task asks how you would make sure it *keeps* doing so after future model, system, and product changes like the one in Task 3.

- How you would turn your Task 2a systematic test into an **automated, standing test**. For example, re-run after every model or pricing-engine change, that fails loudly if a promised discount is no longer delivered in full through the complete waterfall (this is REP 765's "centralised repository" and monitoring recommendation, made concrete).

- How you would **reconcile promised versus actually delivered discounts** across the whole portfolio on an ongoing basis.
- How you would specifically guard against the **order-of-operations bug** re-entering the pricing engine through a future system change.
- How you would manage one risk specific to distributing your product through a third-party channel (e.g., a comparison website or broker) that doesn't apply to direct sales.